JAN-B-1163

30 June 1949

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

BARS, FLINDERS

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Navy 18B5a
15 March 1946

This specification consists of this cover sheet and Navy Department Specification 18B5a dated 15 March 1946, attached hereto, without modification.

Copies of this specification may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.

When a request for this specification is received by a supplying activity, it will be necessary to attach this cover sheet to the pertinent specification before issue.

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18B5a
MAR. 15, 1946
SUPERSEDING
18B5
Aug. 1, 1931

NAVY DEPARTMENT SPECIFICATION

BARS, FLINDERS

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A–1. *Specifications.*—The following Navy Department specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

General Specifications for Inspection of Material.
39P16—Packaging and Packing for Overseas Shipment:
Section V—Boxes, Wood, Cleated, Plywood (JAN–P–105).
Section VI—Boxes, Wood, Nailed (JAN–P–106).
Section VII—Boxes, Wood, Wirebound (JAN–P–107).

A–2. *Other publications.*—The following Bureau of Supplies and Accounts publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Navy Shipment Marking Handbook.

B. TYPES

B–1. Flinders bars, covered by this specification, shall be of the following type:

Type I.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—See section E.

C–2. *Workmanship.*—Workmanship shall be of highest quality and each section shall have an approved black, lusterless finish.

D. REQUIREMENTS COMMON TO ALL TYPES

D–1. See section E.

E. REQUIREMENTS APPLICABLE TO INDIVIDUAL TYPES

E–1. *Composition.*—Flinders bars shall be of highest grade commercial pure soft iron and of purity 99.85 percent iron worked in such a manner as to avoid introduction of mechanical strain due to shear or otherwise.

E–2. *Heat treatment.*—The bars shall be heated slowly and uniformly to a temperature between 1500° and 1600° F. Cooling shall be accomplished in the furnace at a rate not to exceed 350° F. per hour until the temperature reaches 1000° F. when cooling shall then proceed at a rate not to exceed 100° F. per hour until the temperature reaches 400° F., then cooled in still air to approximately atmospheric temperature.

E–3. *Dimensions.*—The flinders bars complete shall be 24 inches in length over-all and shall be comprised of four sections, each of which shall be cylindrical and 2 inches in diameter throughout. One section

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shall be 12 inches in length, one section 6 inches in length, and two sections 3 inches in length.

E–4. *Case.*—The flinders-bar case shall be rectangular in shape, of suitable hardwood, well seasoned and finished; top, sides, and bottom shall be ½ inch thick; the sides of the case and lid shall be dovetailed and secured together by two inside hinges approximately 2 inches from each end; the hinges shall be secured with brass screws; the whole exterior surface of the case shall be varnished; and when the case is closed the top shall be secured to the bottom by a brass hook and screw eye near each end of the case. The case shall be approximately 26 inches in length, 5⅝ inches in width, and 3¼ inches deep, all measurements over all. The inside of the case shall be lined with suitable green felt and shall be divided into two longitudinal sections, into one of which shall fit the flinders bar. In the other there shall be four sections of suitable smooth turned hardwood, similar in lengths to the sections of the flinders bar, but of 1⅞ inches diameter. The grain of these wood spacers shall be approximately parallel to the length of the spacer. The ends of each longitudinal section of the case inside shall have fitted and secured into them a piece of gray felt about ½ inch thick to take up any longitudinal movement or thrust of the bar in transport.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—

F–1a. *Lot.*—For the purposes of sampling, a lot shall consist of not more than 1000 flinders bars in cases offered for delivery at one time.

F–1b. *Sampling procedure for lot acceptance inspection.*—For the inspection specified in paragraph F–2, the Naval inspector shall select representative sample bars in cases in accordance with Table I.

TABLE I.—*Sampling procedure.*

Number of flinders bars in the lot	Number of samples to be selected
1–20	All
21–50	20
Over 50	30

F–1c. *Sampling procedure for chemical analysis.*—From the iron used in the first lot offered for delivery under each contract or order, and thenceforth in at least 1 lot chosen from every 10 lots offered for delivery, the Naval inspector shall take a suitable sample for chemical analysis and shall forward it to a Government laboratory.

F–2. *Inspection.*—Each of the samples selected in accordance with paragraph F–1b shall be subjected to surface inspection by the Naval inspector to determine conformance with the requirements of this specification. If any sample flinders bar and/or case is found to be nonconforming with the requirements of this specification, the entire lot shall be rejected. A rejected lot may be resubmitted for Naval inspection, but only after the manufacturer, subsequent to the first Naval inspection, has inspected the lot 100 percent for the deficiency noted and has removed or repaired all nonconforming bars or cases.

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F–3. *Chemical analysis.*—The sample selected in accordance with paragraph F–1c shall be analyzed to determine conformance with the requirements of paragraph E–1. Acceptance of the first lot shall be withheld pending receipt of a satisfactory test report on the sample representing the lot. Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling and inspection of paragraphs F–1b and F–2 and acceptance shall not be withheld pending receipt of test reports on chemical analyses. However, upon receipt of an unsatisfactory test report on a chemical analysis, the Naval inspector shall select a sample from every subsequent lot offered for delivery and shall submit the sample so selected to the Government laboratory for analysis. Lots shall then be accepted only upon receipt of a satisfactory report on the samples so selected. This additional testing shall be discontinued and lot acceptance returned to the normal basis when 4 successive lots have been accepted.

G. PACKAGING AND PREPARATION FOR SHIPMENT

G–1. *Packing.*—

G–1a. *For domestic shipment.*—Unless otherwise specified, flinders bars, packaged in carrying cases as described in paragraph E–4, shall be packed in cleated plywood boxes, nailed wood boxes, or wirebound boxes, conforming to the requirements of Navy Department Specification 39P16, Section V (JAN–P–105), Section VI (JAN–P–106) and Section VII (JAN–P–107), respectively. The gross weight shall not exceed approximately 150 pounds.

G–1b. *For overseas shipment.*—Unless otherwise specified, flinders bars, packaged in carrying cases as described in paragraph E–4, shall be packed in cleated plywood boxes, nailed wood boxes, or wirebound boxes, conforming to the requirements of Navy Department Specification 39P16, Section V (JAN–P–105), Section VI (JAN–P–106) and Section VII (JAN–P–107), respectively. Shipping containers shall be strapped in conformance with the appendix of the applicable box specification. The gross weight shall not exceed approximately 150 pounds.

G–1c. Where practicable, shipping containers of similar construction shall be of uniform size and shall contain the same number of items of one size and type. Containers shall be designed to fit the contents in a compact manner.

G–2. *Marking.*—In addition to any special marking required by the contract or order, shipping containers shall conform to the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Ordering information.*—Invitations for bids and contracts or orders should specify the following:

Title, number, and date of this specification.
Whether the subject commodity is to be packaged and prepared for domestic or overseas shipment (see section G).

H–2. *Specifications and other publications—where obtainable.*—Copies of Navy Department specifications, Joint Army-Navy specifications and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should

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make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. When requesting specifications, refer to both title and number (or symbol).

H–3. *Patent notice.*—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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